## 0.1 $p$-Groups

**Theorem 0.1.1.** Let $G$ be a group of order $p^n$ for some prime $p$ and $n \in \mathbb{N}$. Then, $G$ has a normal series $\{P_i\}_{i=0}^n$ s.t

$$1 = P_0 \lhd P_1 \lhd \cdots \lhd P_{n-1} \lhd P_n = G$$

where $|P_i| = p^i$ for each $i$.

*Proof.* First, we recall that for any $p$-group $G$, the center $Z(G) \neq 1$. This follows from the class equation:

$$p^n = |G| = |Z(G)| + \sum |G : C_G(g_i)| = |Z(G)| + \sum p^{r_i} \quad (1 \leq r_i \leq n-1).$$

This implies $p$ divides $|Z(G)|$, so $|Z(G)| > 1$.

Now, we use induction on $n$. For $n = 2$, $|G/Z(G)| = p$ or $1$. In either case, $G/Z(G)$ is cyclic, which implies $G$ is Abelian. By Cauchy's Theorem, $G$ has a subgroup of order $p$, and the claim is satisfied.

For $n \geq 2$, suppose the claim is true for groups of order $p^n$. Let $|G| = p^{n+1}$. Since $1 \lneq Z(G)$, by Cauchy's Theorem, there exists a normal subgroup $N \lhd G$ such that $N \leq Z(G)$ and $|N| = p$. Now, $G/N$ is a group of order $p^n$. By the induction hypothesis, $G/N$ has a normal series:

$$1 = N_0/N \lhd N_1/N \lhd \cdots \lhd N_{n-1}/N \lhd N_n/N = G/N.$$

The th Isomorphism Theorem gives $N_{n-1} \lhd G$ and $|N_{n-1}| = p^n$. Thus, by induction, the claim is proved. □

**Lemma 0.1.2.** $G/Z(G)$ is cyclic $\iff$ $G$ is Abelian.